

# Collective-Aware Router Microarchitecture in gem5 Garnet

Multicast, express-link bypass, and tensor all-reduce evaluated against matched baselines.

Chunyu Liu ▪ Boyan Pu

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# Overview

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**01**      **Introduction and Background**

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**02**      **Multicast**

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**03**      **Bypass**

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**04**      **Trace-based Tensor All-Reduce**

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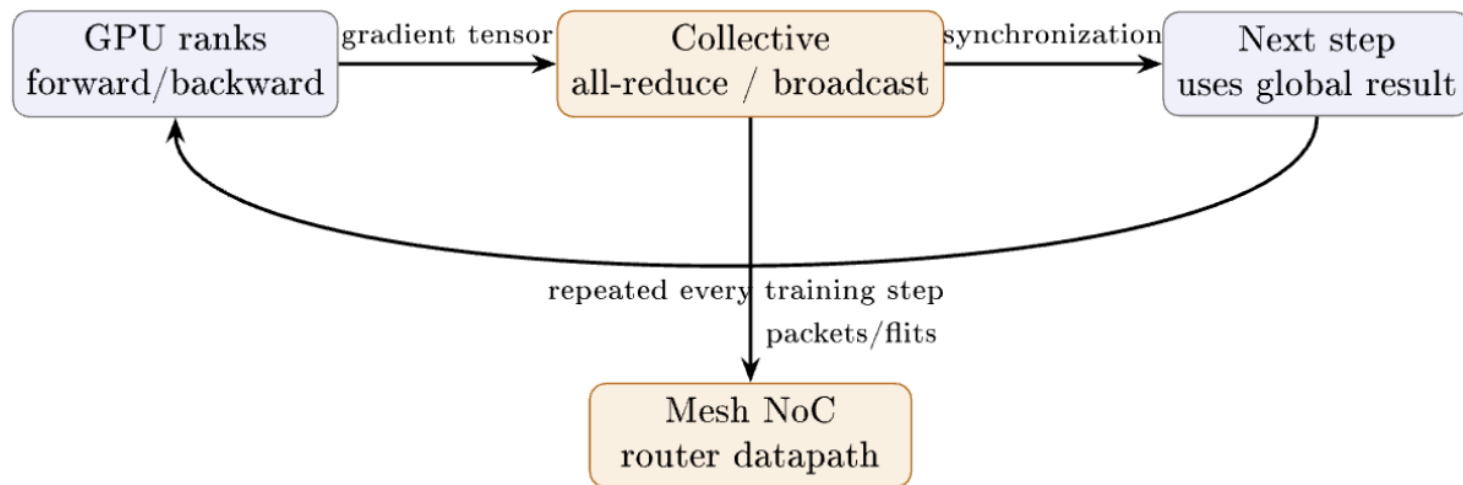
## SECTION 1

# Introduction and Background

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# Research background

AI training moves huge tensors between compute units at every step — so the interconnect is on the critical path.



This report evaluates this layer:  
traffic, latency, and completion correctness.

## 01 Shared-path delivery

**Multicast delivery** shares common paths and replicates only where destinations split.

## 02 Shorter network routes

**Shortcut routing** skips intermediate routers when admission is safe.

## 03 Measured workloads

**Trace-driven replay** maps measured tensor events into network traffic.

**OUR SCOPE** We implement and evaluate all three mechanisms at the network layer in gem5 Garnet.

SECTION 2

# Multicast

# Tree multicast

Send one packet down a tree; copy only where paths split.

## BASELINE

### Replicated unicast

The NI injects one packet per remote destination and records a selected local destination. Shared path prefixes carry the same payload repeatedly.

## PROPOSED

### One bitmap packet

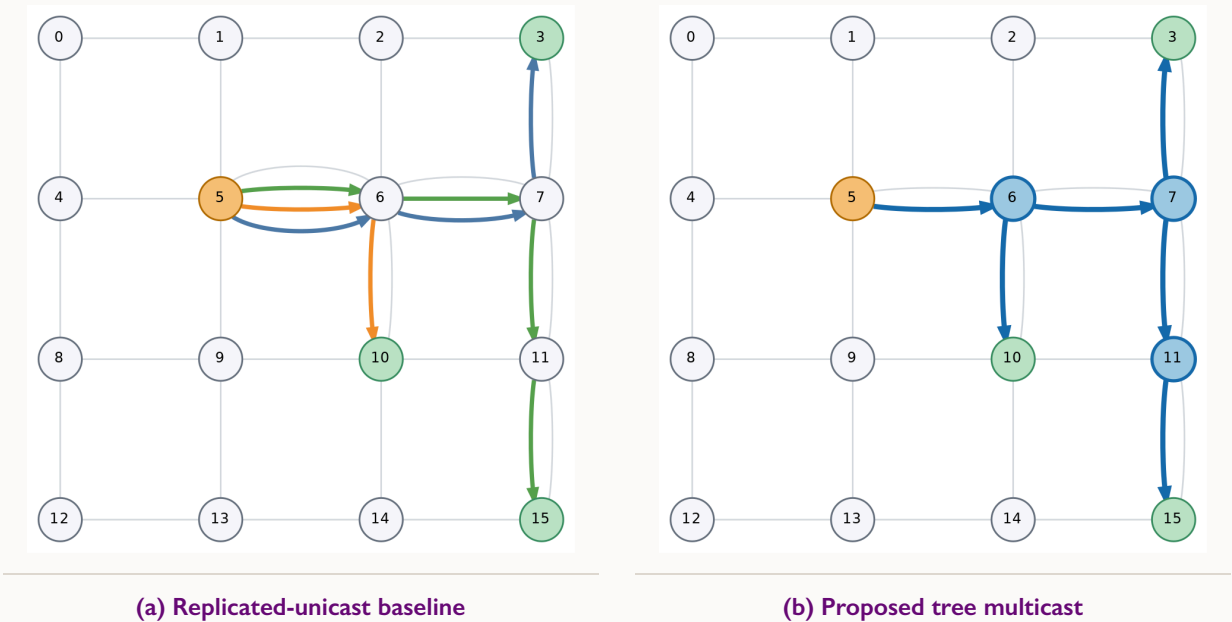
A 64-bit destination bitmap travels in the head flit. Each Router computes only the child branches that still lead to a destination and delivers locally when selected.

## TRADEOFF

### Exact, but synchronized

Fanout is atomic: a flit advances only after every selected output VC has credit. This preserves multi-flit delivery, but a blocked branch can hold the others.

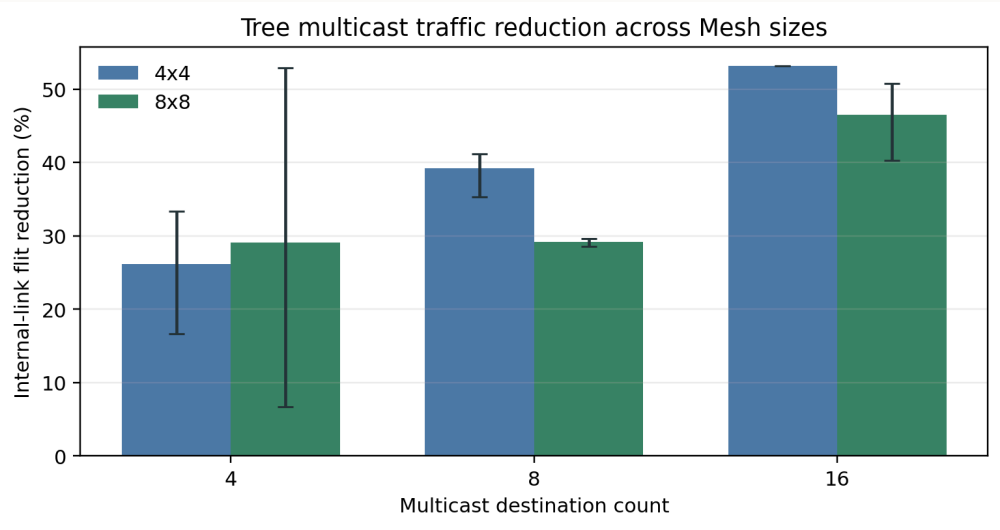
**CORRECTNESS CONTRACT** Every destination receives the complete packet exactly once, in flit order, with no missing or duplicate delivery.



Illustrative 4x4 multicast baseline and proposed path.

# Multicast results

At matched fanout, multicast moves fewer flits on both mesh sizes.



Each bar averages the 1-, 4-, and 16-flit packet cases at that common fanout; error bars show the observed min-max.

| METRIC         | 4x4      | 8x8      |
|----------------|----------|----------|
| Link-flit red. | 39.51%   | 34.95%   |
| Latency        | 3.638×   | 3.472×   |
| Throughput     | +190.89% | +208.34% |

Latency is geometric mean; link-flit reduction and throughput are arithmetic means. Medians: 41.18%/29.63% traffic and 3.588×/2.667× latency.

**READING** Link-flit saving is structural. Latency includes the dedicated Router datapath. Throughput reflects the cost of coordinated fanout.

# Higher fanout, stronger speedup

The bigger the fanout, the bigger the multicast advantage.

-62.41%

link flits at fanout 32  
8x8 scale-out study

11.25x

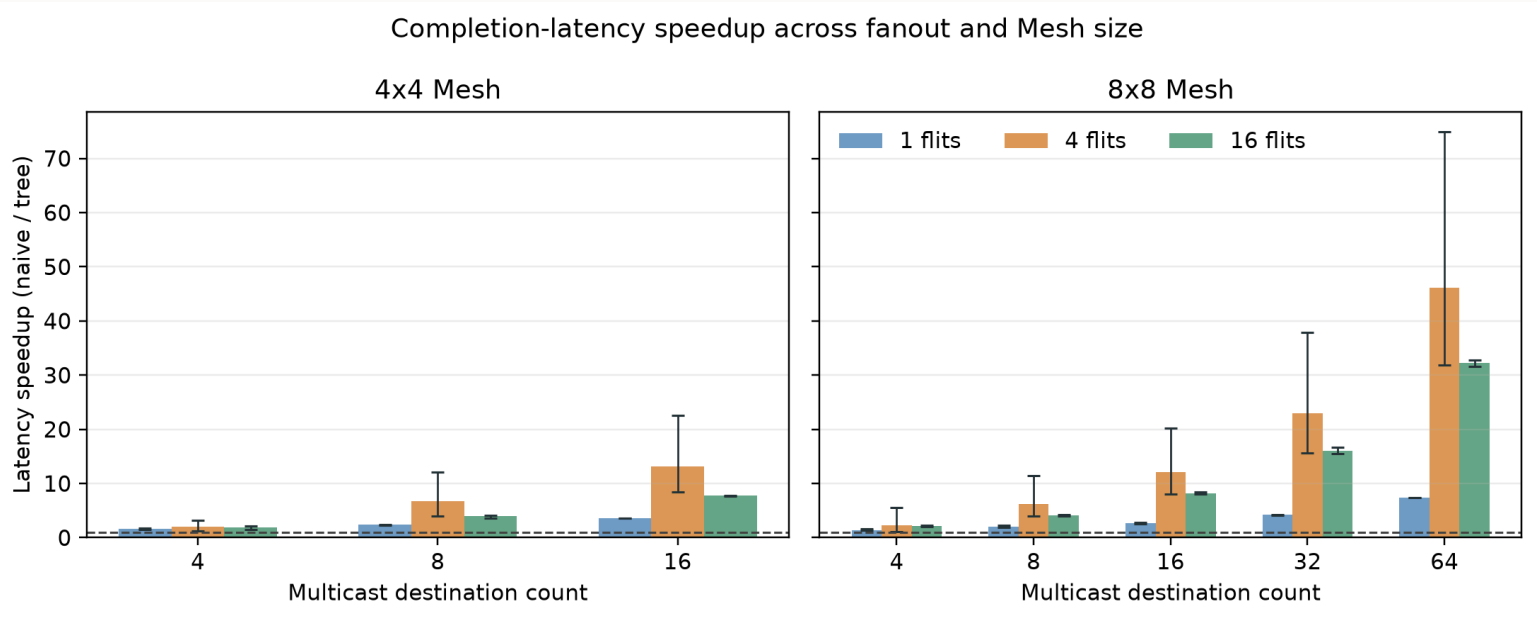
geometric-mean latency speedup  
fanout 32

-75.39%

link flits at fanout 64  
8x8 scale-out study

21.62x

geometric-mean latency speedup  
fanout 64



Scale-out means: +895.54% throughput at fanout 32 and +1,888.29% at fanout 64. These groups are not pooled with the common-fanout mean.

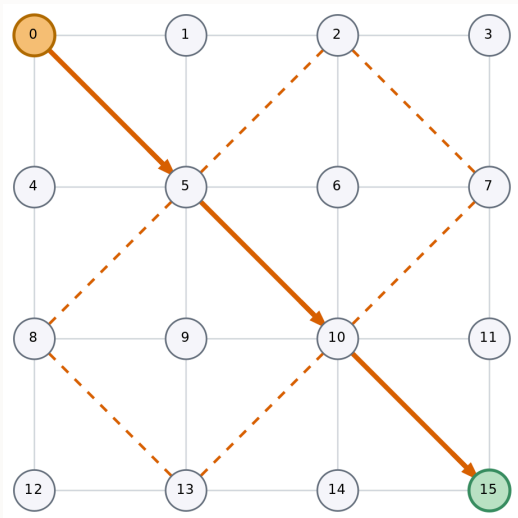
Overall, larger fanout makes multicast speedups more pronounced.



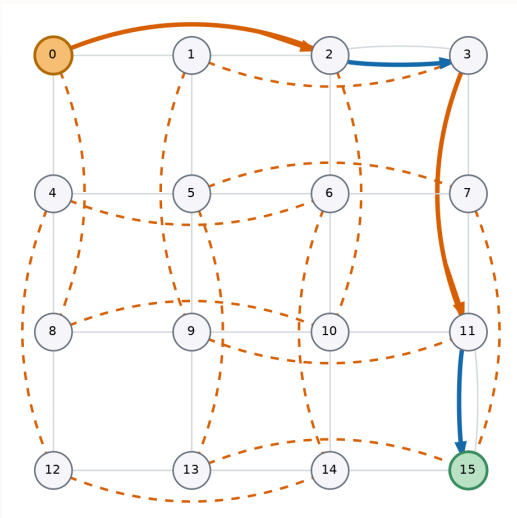
SECTION 3

# Bypass

# Bypass topology



Diagonal



Stride-2 hops while preserving DOR

## MESH BYPASS

### Ordinary Mesh plus shortcuts

Bidirectional stride-2 links are appended; ordinary routers and XY links remain.

## CYCLE-AWARE TABLE

### Multi-hop X then Y

Express latency scales with Manhattan span. An edge is selected only when it lowers the complete remaining path cost.

## BASELINE

### Plain Mesh XY

The comparator has fewer links, ports, VCs, and buffers. This is an added-resource comparison, not iso-area or iso-power.

# Bypass routing

An offline table picks each shortcut; at runtime we only decide whether to take it.

## 1. Price the whole suffix

At every current router and destination, compare ordinary XY with each legal express edge over the complete remaining path.

## 2. Prefer XY on ties

Ordinary cost is link latency plus Router latency; express cost is distance-scaled wire latency plus one Router traversal and the best suffix.

## 3. Keep the phase order

Stride-2 repeats aligned +2 hops within X, then within Y. Diagonal uses X\*, diagonal\*, then Y\*.

## 4. Admit adaptively, within two choices

Ordered Vnets follow the table. Unordered traffic selects express or monotonic XY using local credits, landing pressure, and hotspot state.

```
best_stride_step(current, destination):
    if current == destination:
        return 0
    choices = [ordinary_edge_cost + best_stride_step(xy_next, destination)]
    for legal_bypass in outgoing[current]:
        bypass_cost = wire_latency(legal_bypass) + router_latency
        suffix_cost = best_stride_step(legal_bypass.dst, destination)
        choices += [bypass_cost + suffix_cost]
    return min(choices)  # exact tie: ordinary XY
```

### Diagonal

X\* → diagonal\* → Y\*

### Stride-2

aligned +2 X hops → aligned +2 Y hops

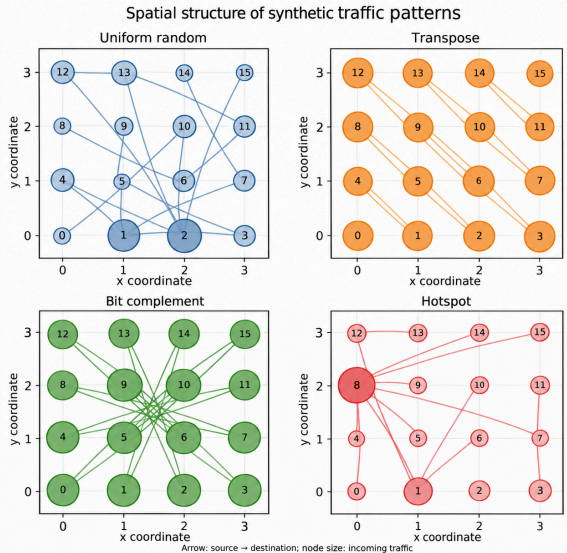
# Bypass results

Across 1,536 matched runs per family, stride-2 scales best on 8×8.

We run 2 mesh sizes × 4 traffic patterns × 4 packet sizes × 16 offered loads × 3 seeds — **1,536 matched cases per bypass family**. Each case runs plain Mesh XY and one bypass arm with the same traffic, packet size, load, seed, and pre-generated source/destination counts, giving 3,072 gem5 runs per family.

| TOPOLOGY | MESH | LATENCY | THROUGHPUT |
|----------|------|---------|------------|
| Diagonal | 4×4  | 1.426×  | +4.27%     |
| Diagonal | 8×8  | 1.165×  | +12.90%    |
| Stride-2 | 4×4  | 1.312×  | +4.37%     |
| Stride-2 | 8×8  | 1.620×  | +39.37%    |

Distance-scaled diagonal and stride-2 · geometric-mean latency · arithmetic-mean throughput.



Uniform random · transpose · bit-complement · hotspot · arrow: source → destination, node size: incoming traffic.

1.620×

stride-2 latency on 8×8

the strongest topology/mesh aggregate

1.426×

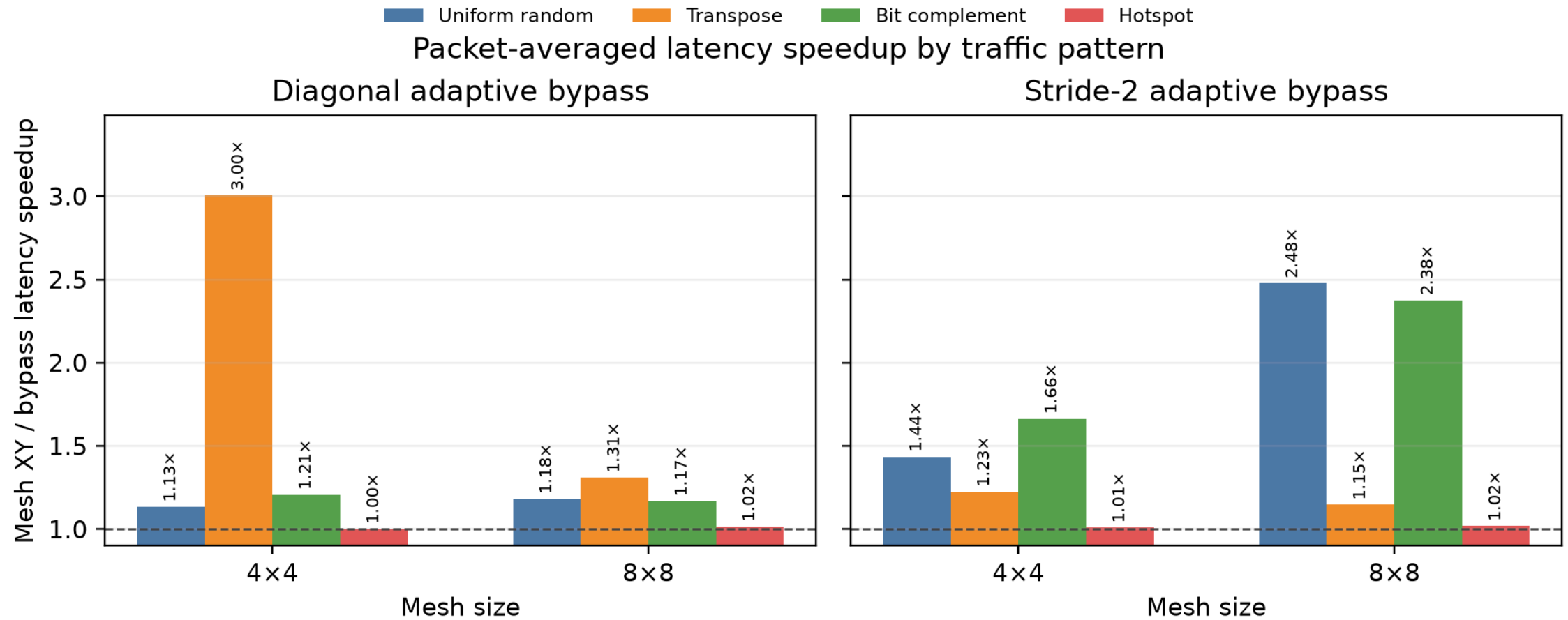
diagonal latency on 4×4

the best diagonal aggregate

**HONEST BOUNDARY** These are workload-matched, added-resource comparisons: the extra links, ports, and buffers are not physically free.

# Latency

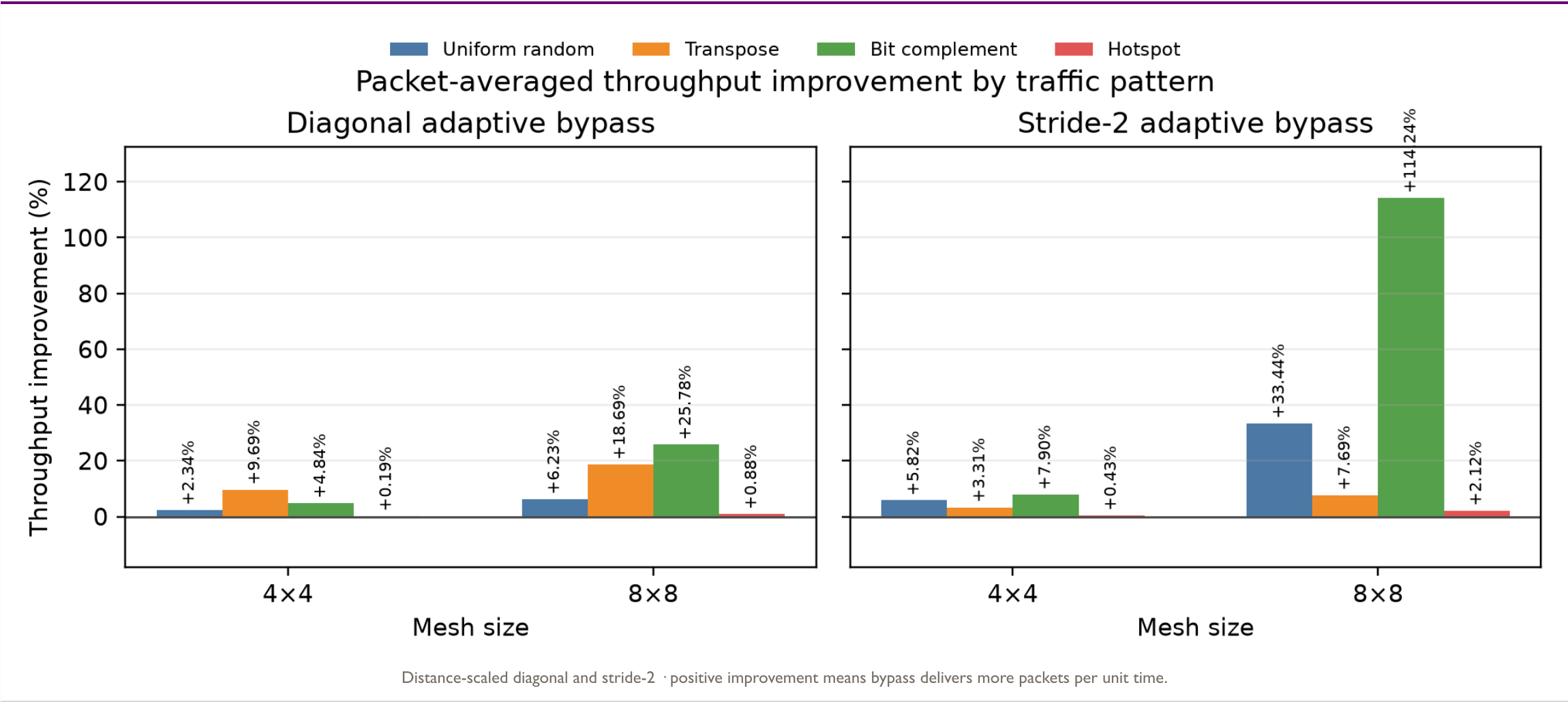
Which traffic patterns actually benefit from the shortcuts?



Distance-scaled diagonal and stride-2 : higher than 1x means bypass finishes before Mesh XY.

# Throughput

Throughput tells the same story: long paths gain the most.

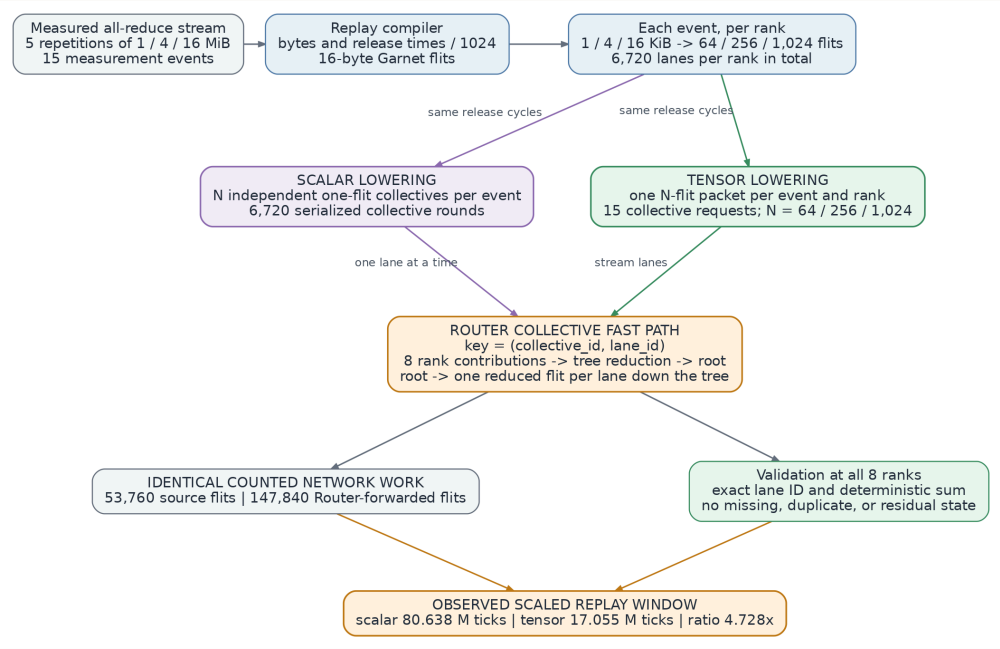


## SECTION 4

# Trace-based Tensor All-Reduce

# HI00 trace replay

Real NCCL events from eight HI00s, replayed two ways in Garnet.



| LOWERING | REQUEST SHAPE                | SCALED WINDOW  |
|----------|------------------------------|----------------|
| Scalar   | many one-flit rounds         | 80.638 M ticks |
| Tensor   | one multi-flit request/event | 17.055 M ticks |

4.728x  
shorter scaled replay window  
same payload and tree work

Trace-based traffic pattern: real NCCL collective events from 8×HI00 become an operation-level Garnet replay; the figure shows the shared work and the two request representations.

**TAKEAWAY** Tensor does not send fewer bytes; it preserves one collective event as one multi-flit request, removing scalar request serialization.



# Division of labor

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## Chunyu Liu

- Tree multicast design and evaluation
- H100 trace capture and compilation
- Replay tooling and tensor experiment statistics
- Report, and slide production

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## Boyan Pu

- Express-link bypass design and evaluation
  - Tensor all-reduce implementation
  - Backpressure and correctness validation
  - Report and slide production
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**Thank you**

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